



Summarizing Recent Research on Predictive Risk Models in Child Welfare

April 2024

INTRODUCTION

Since 2016, the Allegheny County Department of Human Services (DHS) in Pennsylvania has utilized the Allegheny Family Screening Tool (AFST), an algorithm designed to assist child welfare call screening caseworkers in their assessment of general protective service (GPS) referrals regarding potential child maltreatment. Read more about the AFST [here](#). In compliance with Pennsylvania law, GPS referrals can either be screened in for investigation or dismissed at the discretion of the county. The AFST employs administrative data trained to estimate the probability of safety issues so significant that a judge will order a child to be removed from their home within 2 years of the report of suspected abuse or neglect. This information is used alongside other information in the referral process to support call screening case workers in their decision-making process.

The primary objectives of adopting the AFST are to improve the quality of decision making among call screeners by: (1) reducing the frequency of failure to detect conditions that may contribute to future maltreatment; (2) reducing intrusive investigations where there is low likelihood that maltreatment exists; and (3) increasing the consistency of call screeners in their use of data in deciding which GPS reports to investigate.

Although DHS's child welfare call screeners always maintain the discretion to use the AFST risk score as part of a holistic consideration of risk, the inclusion of a data-driven model in decision making has been controversial. It has also attracted extensive research on the tool's effectiveness and how it is perceived by stakeholders. In this paper, we review the research evidence on algorithms in child welfare with a focus on causal analyses of the AFST and similar predictive risk models (PRMs). We first discuss the tools' effects on child welfare decisions. Next, we consider perceived effects, which are often quite different from the actual effects. Finally, we address the need to bridge the gap between perception and reality to reduce concerns that may be unfounded and optimize the utility of the AFST and related tools.

HOW THE AFST WORKS

The initial development of the AFST relied on a PRM constructed from the analysis of nearly 77,000 referrals to child welfare from 2010 to 2014. The analysis used 800 available administrative data features (e.g., prior referrals to child welfare) available at the time referrals were screened to predict downstream outcomes for children. The first version of the algorithm identified features that were most strongly predictive of re-referral and removal; subsequent versions eliminated the re-referral outcome and predicted removal alone. The model predicting re-referrals lacked face validity because high scores could reflect outcomes other than serious abuse and neglect.

Child maltreatment referrals start with a report of suspected abuse or neglect from a source detailing information about the incident and allegations, the alleged victim child(ren), alleged perpetrator(s) and associated individuals. The call screening unit validates this information with the reporting source and against official data sources and adds any additional relevant individuals in the household. Names are cross-checked against DHS's Data Warehouse, and, once the referral has complete data on clients, risk, safety and allegations, call screeners generate a referral risk score from 1 to 20 via the AFST.

Accompanying policies relevant to using the AFST and how it affects the screening decision have been developed and implemented. These policies have changed over time, driven by leadership, but also reflecting input from frontline staff and our ongoing analyses of data. Currently, when the AFST score is 18–20 and at least one child younger than 16 years of age is present in the household, the call screener is alerted that the report has been designated as “high-risk protocol” rather than receiving a numeric score. Likewise, when the AFST score is 1 to 12 and all children are aged 7 years or older, the call screener is notified that the report has been classified as “low-risk protocol.” These protocols default the screening decision in the application to screen-in (if high risk) or screen-out (if low risk), although intake supervisors have discretion to override the defaulted selections and do so regularly. **Figure 1** displays 3 call screener AFST experiences: no protocol, high-risk protocol, and low-risk protocol.

FIGURE 1. Screener Display of AFST Scores and Protocols
(a) No Protocol

Allegheny Family Screening Tool
 Please click the Calculate button to run the algorithm.

Lower Risk
Medium Risk
Higher Risk

10

Last Run By :

Last Run Date :

Algorithm Version Used:
 LASSO v19

The Allegheny Family Screening Tool considers hundreds of data elements and insights from historic referral outcomes to estimate the likelihood of this referral resulting in the need for a child's protective removal from the home within 2 years. It is only intended to help inform call screening decisions, and is not intended for use in investigation or other decision - nor should it be considered a substitute for clinical judgement.

(b) High-Risk Protocol

Allegheny Family Screening Tool

Please click the Calculate button to run the algorithm.

[Calculate Screening Score](#)

		High-Risk Protocol High-Risk and Children Under Age 16 on Referral
Lower Risk	Medium Risk	Higher Risk

Last Run By : 	Last Run Date : 	Algorithm Version Used: LASSO v19
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The Allegheny Family Screening Tool considers hundreds of data elements and insights from historic referral outcomes to estimate the likelihood of this referral resulting in the need for a child's protective removal from the home within 2 years. It is only intended to help inform call screening decisions, and is not intended for use in investigation or other decision - nor should it be considered a substitute for clinical judgement.

(c) Low-Risk Protocol

Allegheny Family Screening Tool

Please click the Calculate button to run the algorithm.

[Calculate Screening Score](#)

Low-Risk Protocol Low-Risk and All Children Age 7+ on Referral		
Lower Risk	Medium Risk	Higher Risk

Last Run By : 	Last Run Date : 	Algorithm Version Used: LASSO v19
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The Allegheny Family Screening Tool considers hundreds of data elements and insights from historic referral outcomes to estimate the likelihood of this referral resulting in the need for a child's protective removal from the home within 2 years. It is only intended to help inform call screening decisions, and is not intended for use in investigation or other decision - nor should it be considered a substitute for clinical judgement.

In all cases, screeners and their supervisors can use the information as guidance but have decision-making discretion. After a decision to investigate has been made, neither the assigned investigator nor any caseworker ever sees the FST score, so it does not affect ultimate assessments of maltreatment or decisions to open a case or remove a child from their home. The FST is designed to complement, not replace, human decision-making at intake where there are more than 80 decisions a day.

Since 2016, when Allegheny County implemented the FST to support child welfare screening decisions, similar systems have been introduced in various jurisdictions across the country. Today, decision aids are being used in several U.S. states and counties, including Los Angeles County, California and Larimer, Arapahoe and Douglas counties, Colorado.

ALTERNATIVES TO PRMS IN CHILD WELFARE AGENCIES

To understand the impacts of predictive risk algorithms in child welfare, it is crucial to consider other methods by which maltreatment screening decisions are made. In the research findings, these other methods represent the status quo. The FST falls into a class of tools conceptualized as “risk assessment tools.” Most existing risk assessment tools used by child welfare agencies are “operator driven” or manual in nature, including the Structured Decision Making (SDM) risk tool—a version of which is implemented in approximately two thirds of states. For states that have not implemented SDM, most have developed structured checklists for assessing risk. Manual tools require information to be gathered and entered by case workers; these tools have several drawbacks, including that they: (1) must be brief enough for workers to complete quickly, so they may fail to incorporate the full range of factors relevant to understanding differences in risk; (2) frequently contain subjective data elements that can lead to biased assessments; and (3) are prone to errors in how they are filled out and completed. In addition, almost none of these tools are locally validated, which means they don’t account for the differences in context and needs that exist in a specific location (i.e., Allegheny County).

Prior to implementation of the FST, Allegheny County did not use any assessment tool. Call screening case workers could access and use historical and cross-sector administrative data on individuals associated with a report of child abuse or neglect. These case workers were required to review relevant information before a screen-in/screen-out decision was made. However, call screeners could not efficiently access, review and make meaning of all available records. In addition, there was inconsistency in decisions by different call screeners.

As a result, call screening case workers screened-in almost half of the lowest-risk-cases and screened out one in four of the highest risk ones.

In 2016, the Children’s Data Network at the University of Southern California received a 2-year research grant to develop a proof-of-concept PRM for California modeled closely on the work in Allegheny County. The goal was to assess whether a PRM built exclusively from child protection records could advance the state’s ongoing efforts to ensure consistent, equitable and sound decision-making at entry to the system.

As part of this research project, attempts were made to not only document the classification accuracy possible through a PRM such as the FST, but also assess how a PRM might improve child welfare practice and supervision beyond what exists today. This assessment included comparing the PRM score with the SDM family risk assessment tool already in use in California. This was critical to the project because it allowed us to document when PRMs could generate additional (or even different) insights about children's risk trajectories. Importantly, it also helped illustrate how those insights from an automated model, on the margins, might ultimately help, harm, or make no difference at all.

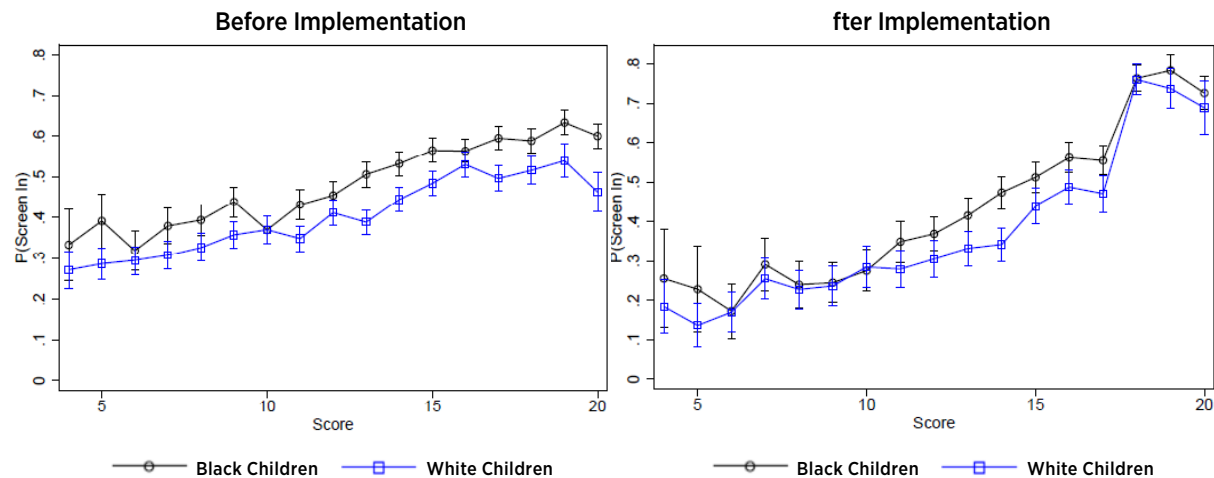
Findings indicated that the PRM was more accurate than the SDM risk tool in identifying children who would have chronic or intensive involvement with the child protection system. In California, researchers compared the accuracy of risk classification decisions from a SDM family risk assessment with PRM results. They found that the top 7% of children identified by the algorithm as "very high risk" were more likely to be placed in foster care than an equivalent-sized group identified as very high risk through the SDM risk assessment tool (74.9% vs. 50.3%), more likely to have 3 additional referrals than the SDM-identified highest risk group (29.5% vs. 12.7%), and more likely to have a substantiated allegation of abuse (33.0% vs. 12.8%). These results suggest that the riskiest referrals flagged by the PRM had higher rates of adverse outcomes than the riskiest referrals identified through the SDM (Putnam-Hornstein et al., 2018). Other white papers, audits and research analyses have also documented serious shortcomings of manual risk assessment tools in terms of both accuracy and implementation (e.g., Bosk, 2018; California State Auditor, 2019; Gillingham, 2011). Reinforcing this point, a recent scoping review found a very thin evidence base for the use of most risk assessment tools (McNellan et al., 2022).

HOW THE FST IS BEING USED AND ITS IMPACT

The following sections detail the main conclusions derived from recent research on the FST and other predictive risk models implemented by child welfare agencies.

The FST changed the composition of investigated referrals.

Rittenhouse et al. (2023) compared screen-in (i.e., investigation) rates by risk score before and after the deployment of the FST. Before FST deployment, scores reflect those that would have been shown to call screeners had the tool been launched. The post-deployment period reflects decisions that were made in the context of an FST score. **Figure 2** shows that the deployment of the FST reduced the probability of screen-in decisions for children in the lowest risk referrals and increased the probability of screen-in decisions among children in the highest risk categories. The introduction of the FST also reduced the racial gap in screen-in rates among higher risk referrals. Prior to implementation, Black children were screened at higher rates than White children, especially for those identified as higher risk. After implementation, the gaps in screen-in rates by race were reduced.

Figure 2. Screen-In Rates by FST Score Before and After FST Deployment for White and Black Children

Note: The left graph displays FST scores prior to FST deployment (January 2013 through June 2016). The right graph displays FST V3 scores (July 2019 through December 2020).

The FST impact on case openings is inconclusive

Among referrals screened in for investigation, the percentage of case openings increased (Rittenhouse et al., 2023). The most plausible explanation for this change is that the FST helped screeners more effectively identify referrals that required further intervention. Goldhaber-Fiebert and Prince (2023) found a 20% decrease in case-opening rates for GPS referrals; however, they also noted that CPS referrals saw a similar relative reduction in case openings, underscoring the uncertainty about interpreting their finding as being driven by the FST rather than environmental or policy changes. Although other results across studies tended to show similar patterns, the case-opening outcomes might be more dependent on choice of analysis timeframe and methodology.

The FST is reducing, not increasing, racial disparities.

Although Figure 2 provides a compelling picture of the changes in screened-in referrals after the introduction of the FST, the results cannot be interpreted as causal unless imposing strong assumptions implicit in before-and-after comparisons. To address these assumptions, Rittenhouse et al. (2023) compared child welfare-related outcomes for Black and White children before and after the algorithm's deployment, using the evolution in the Black-White gap among referrals unaffected by the FST as a comparison. They found that use of the algorithm reduced racial screening disparities across FST scores, although the size and precision of the reduction varied. The FST reduced the racial disparity in screen-in rates for the highest risk referrals by 83%, from 10.6% to 1.8%. The difference based on race in case-opening rates after initial investigation also narrowed. Additionally, the researchers estimated that the FST reduced the Black-White gap in removal rates of screened-in referrals by 73%, from 4.3% to 1.2% (Figure 3).